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ORIGINAL RESEARCH

Segmentation of overlapping leukemic cells in histopathological images using HSV- based watershed transformation

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Abstract
Accurate segmentation of white blood cells (WBCs) is essential for computer-aided diagnosis, as overlapping and densely clustered cells often present significant challenges. This work introduces a hybrid framework for segmentation that proposes a fusion of hue and saturation in the Hue Saturation Value (HSV) domain. Gaussian smoothing, Otsu thresholding, and Morphological refinement is employed to enhance cell contrast and eliminate noise. A marker-based watershed algorithm is subsequently applied for accurate separation of overlapping WBCs. Evaluation on the ALL-IDB2 dataset confirms the method's capability through achieving a Dice Similarity Coefficient(DSC) of 0.8929 and an Intersection over Union (IoU) of 0.8099 to produce well-defined cellular boundaries. The novelty of this study lies in the integrated hue-saturation fusion and marker-based watershed strategy, offering improved boundary localization and reliable segmentation of overlapping WBCs.

Keywords White blood cells (WBCs) · Overlapped cells · Segmentation · HSV color space · Marker-Watershed segmentation

1 Introduction

Acute Lymphoblastic Leukemia (ALL) is a malignant blood and bone marrow condition that comprises uncontrolled proliferation of immature lymphocytes [1]. This dysregulated overproduction of defective lymphocytes interferes with the development of normal blood cells, leading to an array of symptoms such as anaemia, frequent infections, and weakened immunity [2]. ALL is one of the most common childhood hematologic malignancies [3], while it can occur in adults as well, early diagnosis is essential for effective treatment [4]. Morphological analysis of peripheral blood smears is largely based upon the proper identification of ALL and morphologic features of WBCs. However, microscopic inspection of blood smear is labour intensive, subjective and frequently susceptible to diagnostic variations [5]. Research works to automate the analysis of WBCs offers a time efficient and accurate analysis of large number of cellular data [6]. Out of all the image analysis stages, segmentation of WBC images is extremely important because it is the foundation for all the following tasks such as feature extraction, cell classification and subtype differentiation in leukemic diagnosis [7, 8].

Numerous computational methods have been suggested to perform WBC segmentation, Conventional image processing algorithms like thresholding, morphological processing, and watershed transforms have been used extensively because they are easy to interpret and have minimal computational complexity [9]. More recently, deep learning-based architectures like U-Net and its extensions have shown the state-of-the-art performance [10, 11] of learning hierarchical features directly from data. These advanced techniques produce outstanding accuracy but need large annotated datasets and much computational power [12]. As a result, segmentation of leukemic cells is still an active research area with constant efforts towards obtaining robust, accurate, and feasible solutions.

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2 Related work

WBC segmentation from blood smear images has been tackled using different image processing and machine learning methods. Straightforward global thresholding (e.g. Otsu's algorithm) [13] has been employed for segmentation of cells against background, but work poorly when color is non uniform or illumination is inhomogeneous. Color-space approaches have also been widely used; for instance [14], applied the hue channel of HSV color space and triple-thresholding to segment WBCs. In general, converting to HSV or other uniform color spaces is beneficial since it separates chromatic and intensity information, hence color-based thresholding becomes more robust [15].

Region-based and clustering techniques, such as K-means and mean-shift [5], have also been employed to classify pixels into cell and non-cell regions. However, these methods may inaccurately merge bordering cells, particularly in areas of overlap. Marker-controlled watershed segmentation is an established approach for dividing overlapping objects in images. By creating distinct seed markers for each object (normally through the distance transform), watersheds put boundary lines where the regions converge. And this method has been successful in medical image analysis for the separation of overlapping cells [16]. For instance, watershed segmentation has commonly been applied in erythrocyte and nuclei segmentation, and frequently outperforms simple thresholding on overlapping cells [17, 18]. The review paper [11] has demonstrated that morphological and watershed-based techniques are among the most effective approaches for handling overlapping cells in WBC image segmentation.

Deep learning methods (e.g., U-Net) have recently performed outstandingly on cell segmentation [19], but require large annotated datasets and extensive training. Recent advancements in biomedical segmentation highlight the growing adoption of hybrid deep learning and optimized computational models for enhanced diagnostic precision. The paper [20] proposed CNN-TM-GAN, a hybrid convolutional-GAN framework achieving superior edge preservation and boundary accuracy and [21] introduced *ChromoGAT*, integrating U-Net with Graph Attention Networks for precise chromosome segmentation, while [22] developed *VGG1-PCA-PB3C*, combining deep neural features with principal component analysis for efficient leukemia detection. Beyond haematological applications [23], employed a classifier merger strategy to improve Alzheimer's disease diagnosis from MRI, and [24] presented *IFAS*, merging CNNs with morphological segmentation for accurate brain tumour delineation. Then [25] further advanced the field through a zero-shot learning-based-segmentation framework capable of generalizing to thin anatomical structures without retraining.

Although deep and hybrid learning models have advanced biomedical segmentation performance [26], they often rely on large annotated datasets and high computational resources. Classical image processing methods remain valuable for smaller datasets like ALL-IDB2, offering simplicity, interpretability, and reliability. The present work adopts this approach through HSV-based enhancement, morphological refinement, and watershed segmentation to achieve accurate and efficient leukocyte separation.

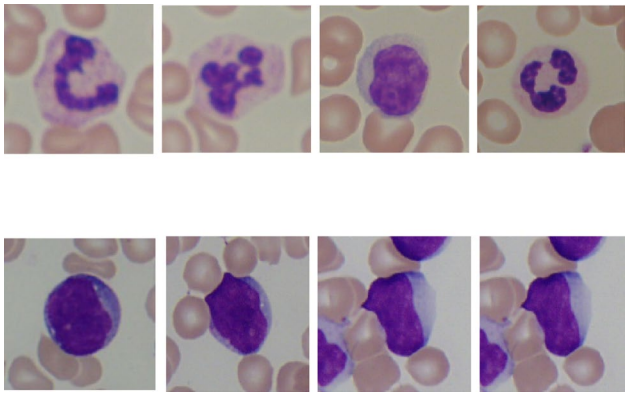

Fig. 1 Sample images from the ALL-IDB2 dataset. Top row: benign lymphocytes from healthy subjects; bottom row: malignant blast cells (ALL)

3 Dataset description

The Acute Lymphoblastic Leukemia Image Database 2 (ALL-IDB2) is a standard public dataset for blood cell image analysis. It consists of 260 single-cell images (130 of normal lymphocytes, 130 of ALL blast cells) taken from blood smear microscopy images [27]. The images were captured by the Università degli Studi di Milano using an optical microscope with a Canon PowerShot camera, and are in JPG format (24-bit color). ALL-IDB2 was generated by cropping regions of interest from the original ALL-IDB1 images, so each image contains one leukocyte (of either class) possibly along with some other blood elements [28, 29]. The color properties (hue and saturation) of ALL-IDB2 images are similar to the originals, although the image sizes vary [27]. Sample images from the dataset are shown in Fig. 1.

Each ALL-IDB2 image is relatively large (originally up to 2592 × 1944 pixels [28]). In practice we resize or process patches if needed, but here we operate on full-size images to preserve details. Importantly, while most images contain a single prominent WBC, some images may include multiple white cells or other interfering cells in the background. These cases create clusters of overlapping cell regions, which pose a difficulty for segmentation. Figure 1 (bottom

row) illustrates some of these complex cases. The varied cell morphology and presence of neighboring cells make segmentation nontrivial [27]. Our goal is to reliably isolate the target leukocyte region(s) even when cells overlap, using only classic image processing methods.

4 Methodology

This study employs a systematic image processing framework designed to segment overlapping WBCs in microscopic smear images from the ALL-IDB2 dataset. The algorithmic workflow followed in this study is outlined in Table 1, which provides a concise representation of the complete segmentation procedure and a detailed explanation of each processing step is presented from Sect. 4.1 onward.

4.1 HSV color space and channel combination

The input microscopic images are represented in the RGB color model as shown in Eq. (1). However, RGB mixes chromatic and intensity information, making it sensitive to illumination. To improve robustness, we transform images into the Hue-Saturation-Value (HSV) model defined in Eq. (2), where hue and saturation capture chromatic content independent of brightness.

Let $R, G, B \in [0,1]$, Define:

$$C_{max} = \max(R, G, B), \; C_{min} = \min(R, G, B),$$

$$\Delta = C_{max} - C_{min} \quad (1) \tag{1}$$

Table 1 Algorithmic representation of the proposed segmentation method

Input: Microscopic blood smear image from the ALL-IDB2 dataset containing overlapping WBCs.
Output: Segment image with clearly separated WBCs and labeled cell instances
Step1: Convert the RGB image into HSV color space to separate chromatic and luminance components.
Step 1.1: Extract Hue and Saturation channels highlighting leukocyte chroma.
Step 1.2: Combine normalized H and S channels as for color enhancement.
Step2: Apply Gaussian smoothing to suppress noise.
Step3: Perform Otsu's global thresholding.
Step4: Perform morphological opening (erosion followed by dilation) to eliminate small artifacts.
Step5: Compute the Euclidean Distance Transform (EDT).
Step6: Generate foreground markers by thresholding the distance map at *0.5*max(EDT)*
Step7: Derive the gradient magnitude image.
Step8: Apply marker-controlled watershed on the gradient image.
Step9: Overlay watershed ridges on the original image.

The HSV channels are then defined as:

$$H = \begin{cases} \frac{1}{6}\left(\frac{G-B}{\Delta} \; mod6\right), & C_{max} = R, \\ \frac{1}{6}\left(\frac{B-R}{\Delta} + 2\right), & C_{max} = G, \\ \frac{1}{6}\left(\frac{R-G}{\Delta} + 4\right), & C_{max} = B \\ 0, & \Delta = 0 \end{cases}$$

$$S \begin{cases} 0, & C_{max} = 0 \\ [6pt] \frac{\Delta}{C_{max}}, & otherwise \end{cases}$$

$$V = C_{max} \tag{2}$$

Since leukocyte nuclei are distinct in hue and saturation, we construct a weighted grey scale image as seen in Eq. (3)

$$I = 0.6H + 0.4S \tag{3}$$

4.2 Gaussian smoothing

Before thresholding, the combined image *I* is smoothed to reduce noise and small texture variations. We apply a Gaussian blur (e.g., 5 × 5 kernel with σ determined automatically). Gaussian smoothing effectively suppresses high-frequency noise (small speckles), making the subsequent binary segmentation more robust. The Gaussian filter defined in Eq. (4) has the property of reducing the influence of any spurious bright or dark pixels, without significantly altering the overall cell shape.

$$G_\sigma(x, y) \; \frac{1}{2\pi\sigma^2} exp\left(-\frac{x^2 + y^2}{2\sigma^2}\right), \tag{4}$$

Then pre-smoothing, which is a common pre-processing step prior to thresholding in image segmentation has employed and the smoothed image is defined in Eq. (5):

$$I_\sigma = I * G_\sigma \tag{5}$$

Where '*' denotes convolution.

4.3 Otsu's thresholding

Segmentation begins with global thresholding of the smoothed image. Otsu's algorithm selects the threshold *t** that maximizes the between-class variance. For histogram probabilities *p(i)*, class probabilities and means are defined in Eqs. (6), (7) and (8):

$$\omega_0(t) = \sum_{i=0}^{t} p(i), \quad \omega_1(t) = \sum_{i=t+1}^{L-1} p(i) \tag{6}$$

$$\mu_0(t) = \frac{1}{\omega_0(t)} \sum_{i=0}^{t} i\,p(i), \quad \mu_1(t) = \frac{1}{\omega_1(t)} \sum_{i=t+1}^{L-1} i\,p(i) \tag{7}$$

$$\sigma_b^2(t) = \omega_0(t)\,\omega_1(t)\,((\mu_0(t) - \mu_1(t))^2 \tag{8}$$

And the optimal threshold is defined in Eq. (9), which produces a binary mask of potential leukocyte regions:

$$t^* = argmax_t\, \sigma_b^2(t) \tag{9}$$

4.4 Morphological opening

The binary mask from thresholding had small isolated noise pixels (salt) or tiny holes. To clean up these artifacts, we performed a morphological opening operation: first an erosion, then a dilation, using an appropriate structuring element. Specifically, we use an elliptical (disk-like) structuring element of size about 13 × 13 pixels. Morphological opening has the effect of removing small objects (smaller than the structuring element) and smoothing object contours while preserving the shape of larger regions. In our context, opening removes small white noise and disconnects very narrow bridges, without eroding the bulk of the actual cell region. After opening, the mask contains only the "sure" foreground: the region(s) where high confidence cells, free of tiny artifacts. Any large foreground gaps (holes) inside cells would also be eliminated by this process.

4.5 Distance transform and foreground markers

To separate overlapping cells, we employed the distance transform method. The distance transform of a binary mask assigns each foreground pixel a value equal to its distance to the nearest background pixel as shown in Eqs. (10),

$$D(p) = \mathrm{mind}(p, q), \quad d = Eucledian\ distance \tag{10}$$

Intuitively, central points of each blob have larger distance values. We compute the Euclidean distance transform on the opened mask. Peaks in this distance map correspond to the centers of candidate cell objects. We then threshold the distance image to extract a *sure foreground* region: pixels whose distance value exceeds a fraction (e.g. 0.5) of the maximum distance. This yields a binary image of peaks that are far from any boundary. By construction, each connected component is well within a single cell. We label these components with unique integer markers using connected-component analysis. These markers will serve as seeds for watersheds.

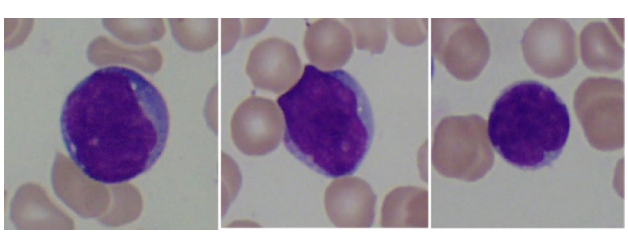

Fig. 2 Raw images from ALL-IDB2

4.6 Marker-controlled watershed segmentation

Finally, we apply the watershed transform using the gradient magnitude of the smoothed image as the elevation map. Given gradient image E and markers M, the watershed divides the image domain into disjoint regions $\{C_1, C_2, \ldots, C_k\}$, where:

- Each region C_i contains exactly one marker $m_i \in M_I$.
- Boundaries are placed along the ridges of E.

This ensures that overlapping leukocytes are separated into distinct segments, with boundaries drawn at points of maximum gradient (cell-cell contacts).

5 Result and discussion

The segmentation work flow described in the methodology was applied to the ALL-IDB2 dataset. To demonstrate the effectiveness of the approach, we present few leukocyte images and illustrate the intermediate outputs at each stage of the workflow. The raw input images from ALL-IDB2 dataset given below in Fig. 2. shows a stained peripheral blood smear containing WBCs(Leukocytes) and RBCs (Erythrocytes). The leukocyte of interest exhibits strong nuclear staining with overlapping cell boundaries.

After converting the RGB image of those raw input images in Fig. 2. to HSV, the hue h ? [0,1] and saturation s ? [0,1] channels are normalized and fused as $I = 0.6H + 0.4S$. In the resulting grey scale, the leukocyte nucleus appears brighter than the erythrocyte in the background due to its distinct chroma. This enhances leukocyte background contrast while suppressing illumination bias from the value channel. Then 5*5 Gaussian blur is applied to I and rescaled to 8-bit as shown in Fig. 3. Smoothing removes high-frequency stain noise and small texture from RBCs without materially shifting leukocyte edges, producing a cleaner histogram for Otsu's thresholding.

Otsu's method selects an image-specific threshold t^* on the blurred grayscale and binarizes it (binary = $I_{blur} > t*$). The leukocyte region forms a contiguous white foreground while most background and RBCs are supressed to black. This mask defines the initial cell candidate region as seen in Fig. 4.

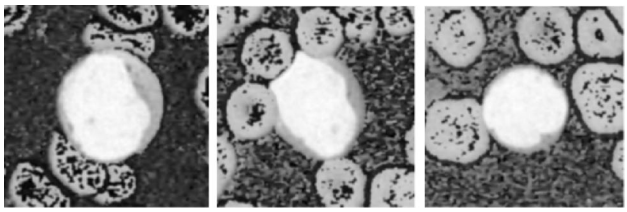

Fig. 3 Images after Gaussian smoothing

Fig. 4 Images after Otsu's thresholding

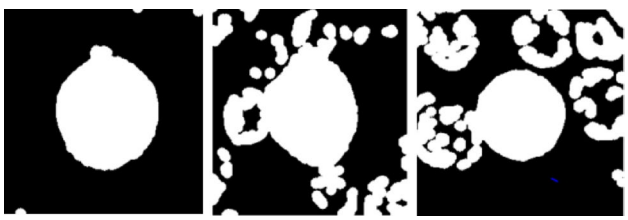

Fig. 5 Images after morphological opening

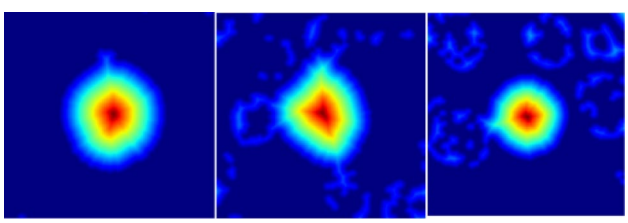

Fig. 6 Euclidean distance transform

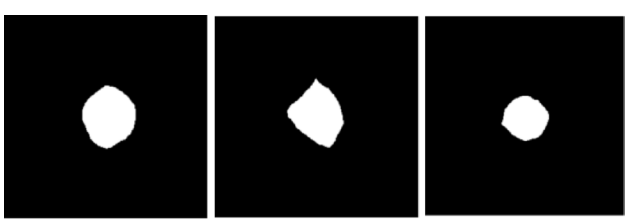

Fig. 7 Sure-Foreground

Then it undergoes morphological opening, which is an elliptical structuring element (size 13 *13) removes small, idicelated specks and thins spurious connections while preserving the leukocyte blob. The opened mask as seen in Fig. 5. is smoother along the nuclear contour and free from tiny artifacts that would otherwise create false seeds. After opening, Euclidean Distance Transform (EDT) of the opened mask peaks, at blob interior were found as seen in Fig. 6. Thresholding at 0.5* max (EDT) yields the

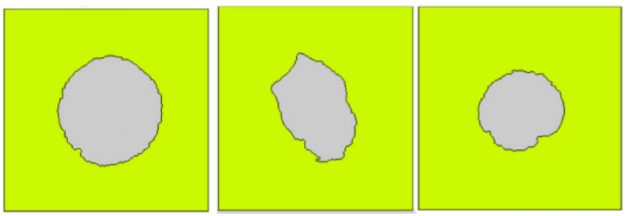

Fig. 8 Watershed markers

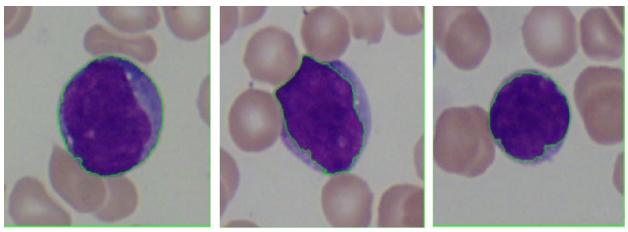

Fig. 9 Watershed contours

sure-foreground, which compact seed regions centered within each leukocyte (white), far from borders. In overlapped cases, multiple interior peaks (and thus multiple seeds) are produced were crucial for later separation, hence dilating the opened mask creates a sure-background as seen in Fig. 7. Subtracting the sure-background from this background produces the unknown band, a thin ring at object borders and between overlapping cells– marking pixels whose final ownership must be decided by watershed. Visually this appears as a dark ring surrounding the bright seed.

Then the connected components on the sure-foreground give integer markers (incremented by + 1, then background is 1). Pixels in the unknown band are set to 0, forcing the watershed to resolve them. Each interior seed corresponds to a putative cell; in multi-cell clusters, there are two or more distinct labels. Hence the segmentation contours were obtained by applying marker-controlled watershed on the gradient image as seen in Fig. 8.

The final result given in Fig. 9 shows the original image with the watershed-derived boundaries overlaid in green. The leukocyte of interest is segmented from surrounding blood cells, and overlapping structures are accurately separated. This confirms the robustness of the proposed method for handling complex smear images with overlapping WBCs.

The experimental results across multiple images in the ALL-IDB2 dataset are consistently accurate delineation of leukocyte(WBC) boundaries. The use of HSV transformation, channel combination of H & S, Otsu's thresholding, distance transform, and marker-controlled watershed allowed for effective separation of overlapped cells, which is essential for subsequent classification tasks.

6 Validation of result

To quantitatively evaluate the performance of the proposed segmentation approach, we employed two widely used overlap-based metrics: the 'Dice Similarity Coefficient(DSC)' and the 'Intersection over Union (IoU)', also known as Jaccard Index. The DSC is defined in Eqs. (11),

$$DSC = \frac{2|X \cap Y|}{|X| + |Y|} \tag{11}$$

where X represents the set of pixels in the predicted segmentation and Y represents the set of pixels in the annotated white blood cell images. The value ranges from 0 to 1, where 1 indicates perfect agreement.

The IoU metric is expressed as shown in Eqs. (12),

$$IoU = \frac{|X \cap Y|}{|X \cup Y|} \tag{12}$$

Which also varies between 0 and 1, with higher values denoting greater overlap between the prediction and annotated mask. For validation, a subset of annotated white blood cell images manually developed was used as ground truth, which was compared with the segmentation results generated by the watershed-based method. Table 2 summarizes the obtained DSC and IoU scores for representative samples.

The experiments show that the suggested segmentation algorithm has a high level of agreement with ground truth annotations. The average DSC value was 0.8929 and the average IoU value was 0.8099, validating the robustness of the methodology for identifying leukemic cells, even in difficult cases with abnormal cell boundaries or overlapping structures. The outcomes are in line with high segmentation performance demonstrated in similar studies. For example [30], obtained DSC ranges between 86% and 97.3% for WBC segmentation on various datasets using RGB operations and Fourier based enhancements. Segmentation based on deep learning architectures like VGG, UNet yielded a DSC of 94.4% and IoU of 91.5%, which is state-of-the-art accuracy. These standards imply that DSC score greater than 0.85 and IoU greater than 0.80 are considered to be highly accurate for WBC segmentation in biomedical imaging. Our attained scores DSC $\approx$ 0.89 and IoU $\approx$ 0.81 are thus comparable and found to be better to the state-of-the-art approaches.

Table 2 DSC and IoU scores for representative samples

Image	DSC	IoU
Im001	0.9269	0.8638
Im002	0.8994	0.8172
Im003	0.8231	0.6994
Im004	0.8184	0.6926
Im005	0.8240	0.7007
Im006	0.8956	0.8109
Im007	0.9355	0.8789
Im008	0.9152	0.8437
Im009	0.9452	0.8960
Im010	0.9453	0.8963

7 Conclusion

In this work, an automated pipeline of segmentation was designed to separate overlapping white blood cells in Acute Lymphoblastic Leukemia (ALL) smear images from the ALL-IDB2 dataset. This research work introduces novel approach in considering the hue and saturation fusion in the HSV domain by utilizing color-space transformation, Gaussian smoothing, Otsu thresholding, morphological refinement, distance transformation, and marker-controlled watershed, the approach successfully defined leukemic cells with a low degree of human intervention. Validation with manually annotated ground truth illustrated excellent performance with average DSC and IoU values of 0.8929 and 0.8099, respectively, establishing the accuracy and stability of the process. The novel approach presented in this research work is therefore a robust, interpretable, and economically feasible solution for leukocyte segmentation that can be used as a precursor to subsequent tasks including feature extraction, subtype classification, and clinical decision support in leukemia diagnosis. In future work, the segmentation framework can be extended using Optical Scanning Holography (OSH) to capture three-dimensional structural information of leukocytes from microscopic smear samples. Incorporating holographic depth features alongside traditional spatial cues could enable more precise delineation of overlapping and irregularly shaped cells.

Declarations

Conflict of interest The authors would like to declare that they have no competing interests and this research work didn't receive any grant from funding agencies.

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